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Pressure response using wavelet analysis in the process of hydraulic fracturing: Numerical simulation and field case

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ABSTRACT

Diagnosis of hydraulic fracturing is of great significance for field treatment guidance. The real-time analysis of the pressure curve is helpful to evaluate the hydraulic fracture status, and provide guidance for the adjustment of the fracturing treatment. However, the response pattern of hydraulic fracture propagation and fluid flow in the pressure curve is still unclear and difficult to capture. In this paper, the 2D hydraulic fracturing model was built to investigate the effects of different stress differences, elastic modulus, and variable rate. Simultaneously, the wavelet analysis method was introduced to observe its response characteristics. Taking the peak energy of the wavelet as the standard, the db5 wavelet basis is selected as the optimal wavelet basis and the first decomposition level is the optimal recognition level. There is a positive correlation between the pressure drop value and the wavelet peak energy under different stress difference and elastic modulus. In addition, we compared the pressure drop ratio and the wavelet energy ratio found that the wavelet energy can greatly amplify the impact of pressure fluctuations. Comparing the wavelet energy of pressure and rate can eliminate the influence of rate fluctuation on pressure. Finally, the field treatment curve was decomposed and explained by wavelet analysis, then it is compared with the results of microseismic interpretation for the first time. This paper can provide further reference for the interpretation of hydraulic fracturing curves.

Credit author statement

Xiaodong Hu: Funding acquisition, Supervision, Conceptualization, Writing-Review and Editing. **Guopeng Huang:** Software, data processing, Writing-Review and Editing. **Fujian Zhou:** Supervision, Writing-Review and Editing. **Yang Qiu:** Data Curation, Writing polish. **Xiaoyan Gou:** Field data processing. **Chao Chen:** Writing polish.

1. Introduction

With the development of unconventional resources exploration, hydraulic fracturing technology has become the most economical and effective way to stimulating production (Dong et al., 2021; Mahmoodi et al., 2019; Ren et al., 2020; Zhang et al., 2021). Thus, understanding the process of hydraulic fracturing is important for evaluating the effect of hydraulic fracturing, and can provide beneficial guidance for subsequent treatment and production. Due to the influence of various factors such as formation factors and treatment factors, there are many abnormal fluctuations on the treatment curve that generally closely related to the propagation of hydraulic fracture and the adjustment of treatment strategies (Guo et al., 2014). Capturing these abnormal fluctuations is significant for understanding the fracturing process.

In recent years, many scholars have conducted related research based on the response mode on the pressure curve. Nolte (1986) explained the pressure response during hydraulic fracturing based on PKN, KGD, and radial model. The response is divided into four types based on the relationship between net pressure and time: when the slope of the curve is a relatively small positive value represents the normal propagation of the fracture; when the slope of the curve is 1 represents sand plugging; when the curve drops rapidly represents an increase in the height of the fracture. However, because this method is based on assumptions such as constant rate and viscosity, it has great limitations in the actual application process (Guo et al., 1994). Kim and Yilin Wang (2014) considered the influence of multiple complex factors (friction, temperature, pressure, etc.) on the bottom hole treatment pressure on the basis of Nolte-Smith et al., and accurately calculated the bottom hole net pressure. Pirayesh et al. (2015) based on Nolte-Smith method believed that fracture propagation is an intermittent process, and

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proposed the Moving Reference Point (MRP) method. Real-time diagnosis of hydraulic fracturing by discriminating e value versus time, such as normal propagation of fracture, height increase, sand out, etc. Wigwe et al. (2018) further developed the MRP method to identify natural fractures in shale formations and determine the length of hydraulic fractures.

The pressure signal during hydraulic fracturing is affected by a variety of complex factors. The signal is non-stationary and transient, and contains different information at different frequencies (Unal et al., 2018). Decomposing the pressure signal into different frequency bands can help us to understand the response characteristics and laws of the pressure signal. And the wavelet analysis is undoubtedly the most suitable tool (Daubechies, 1990). Wavelet analysis uses wavelet functions to convert signals in the time domain to different frequency bands. Compared with Fourier transform and short-time Fourier transform, wavelet analysis has good time and frequency localization analysis capabilities (Fan et al., 2020; Kumar and Foufoula-Georgiou, 1997). Generally, wavelet analysis can be divided into two types: continuous wavelet transform and discrete wavelet transform (Unal et al., 2019b). Continuous wavelet transform is suitable for signals that are continuous in time, and discrete wavelet transform is suitable for discrete signals, such as pressure signals. Since Morlet et al. (1982) first applied wavelet analysis to the petroleum industry, many scholars have applied wavelet analysis to many aspects of the petroleum industry (Guan et al., 2004; Panda et al., 2000; Soliman et al., 2001). Among them, Unal et al. (2018, 2019a, 2019b) used wavelet analysis for the first time in the analysis and interpretation of hydraulic fracturing treatment curves. Through discrete wavelet transformation of the pressure curve, the curve of wavelet detail coefficients over time was obtained、analyzed and explained. The sand out was diagnosed in advance and verified by the MRP (Moving Reference Point) method. Unal et al. (2019a) introduced the energy concept proposed by Burrus et al. (1998). By comparing the energy relationship between pressure and rate, the types of events in the hydraulic fracturing treatment process are divided into two types: Rate-relate and Rock-relate; Based on Unal et al. (2019a) work, Awad et al. (2020) gave a further explanation on the choice of wavelet basis and optimal decomposition level. However, previous studies on pressure response are directly based on field data for analysis, and the actual situation in the well is more complicated, resulting in unclear changes in many wavelet response parameters. Therefore, in order to describe and interpret the pressure signal more accurately, it is necessary to clarify the influence law of the formation parameters on the wavelet response, and to clarify the interaction relationship between the treatment parameters (rate) and the pressure signal in the time-frequency domain.

The two-dimensional (2D) hydraulic fracturing model was established by the finite element method, which to simulates the response process of the injection pressure during the hydraulic fracturing process, and introduces wavelet analysis to analyze the injection pressure signal. Meanwhile, the wavelet response characteristics under different paraments (stress difference, elastic modulus, and rate) are investigated. Finally, the wavelet analysis of the field treatment curve is compared with the microseismic interpretation results, which verifies the feasibility of the wavelet analysis for diagnosis in the hydraulic fracturing process.

2. Methodology

2.1. Hydraulic fracturing model

(1) Mathematical model

The reservoir rock is assumed as a porous medium material, including rock solid particles, pore structure and fluids existing in the pore structure. The rock solid particles and fluids are incompressible (Systèmes, 2014). Through coupling the flow of fracturing fluid and the dynamic deformation of rock, the propagation process of hydraulic fractures is realized. The stress balance equation and fluid continuity equation are given by Eq (1) and Eq (2) respectively:

Stress balance equation:

$$\int_V (\bar{\sigma} - u_w I) : \delta\dot{\varepsilon}\delta dV = \int_S t\delta v dS + \int_V f\delta v dV + \int_V n\rho_w g\delta v dV \tag{1}$$

Continuity equation:

$$\frac{d}{dt}\left(\int_V \frac{\rho_w}{\rho_w^0} n dV\right) = -\int_S \frac{\rho_w}{\rho_w^0} n\mathbf{n}\cdot v_w dS \tag{2}$$

Which, $\delta v$is the virtual velocity field, m/s; $\delta\dot{\varepsilon}$is the virtual deformation rate, s^{-1}; $\bar{\sigma}$is the Cauchy effective stress, Pa; $t$is the surface external force per unit area, N/m^2; $f$is the volumetric force, N/m^3; $\rho_w$is the fluid density, Kg/m^3; $\rho_w^0$is the reference density of the fluid; $g$is the acceleration of gravity, m/s^2; $v_w$is the seepage velocity, m/s; $n$is the porosity, dimensionless; $\mathbf{n}$is the normal direction outside the surface.

The pore pressure cohesive element is used to describe the process of hydraulic fracture initiation and propagation. The maximum principal stress criterion is used to describe the fracture initiation process. When the tensile stress in any direction of the element reaches its tensile strength, the element will begin to fail, the expression is described by Eq (3):

$$\max\left\{\frac{\langle t_n\rangle}{t_n^0}\right\}^2, \left\{\frac{\langle t_s\rangle}{t_s^0}\right\}^2, \left\{\frac{\langle t_t\rangle}{t_t^0}\right\}^2 = 1 \tag{3}$$

The stiffness degradation is used to describe the damage evolution process of the element, it can be expressed as (Wang et al., 2020):

$$\begin{aligned} t_n &= \begin{cases} (1-D)\overline{t_n}, \overline{t_n} \geq 0 \\ \overline{t_n}, \overline{t_n} < 0 \end{cases} \\ t_s &= (1-D)\overline{t_s} \\ t_t &= (1-D)\overline{t_t} \end{aligned} \tag{4}$$

Where $\overline{t_n}$、$\overline{t_s}$and $\overline{t_t}$ respectively are the cohesive element normal, first tangential and second tangential stresses predicted under the current strain according to the undamaged frontal elastic traction-separation criterion, Pa. t_n、t_s and t_t respectively correspond to the stress in the three directions, Pa.

The fluid flow in the fracture can be divided into tangential flow along the fracture and normal flow perpendicular to the fracture. The tangential flow can be described by Newtonian flow, is given by Eq (5), and the normal flow can be described by the leak-off coefficient of the cohesive element, is given by Eq (6) (Guo et al., 2015; Systèmes, 2014):

$$\mathbf{q} = -\frac{w^3}{12\mu}\nabla p \tag{5}$$

Where $\mathbf{q}$is the volume flow vector in the cohesive element, m^3/s, $w$is the size of the cohesive element opening, m, $\mu$is the cohesive element viscosity coefficient, Pa•s; $p$is the fluid pressure in the cohesive element, MPa.

$$\left\{\begin{array}{l} q_t = c_t(p_i - p_t) \\ q_b = c_b(p_i - p_b) \end{array}\right\} \tag{6}$$

Which, q_t、$q_b$are the seepage flow rate of fluid on the upper and lower surfaces of the fracture, m^3/s;c_t、 $c_b$are the leak-off coefficients of fluid on the upper and lower surfaces of the fracture; p_t、$p_b$are the pore pressures of fluid on the upper and lower surfaces of the fracture, MPa; $p_i$are the fluid pressures on the middle surface of the Cohesive unit, MPa.

(2) Physical Model

A two-dimensional (2D) model with natural fracture is established in

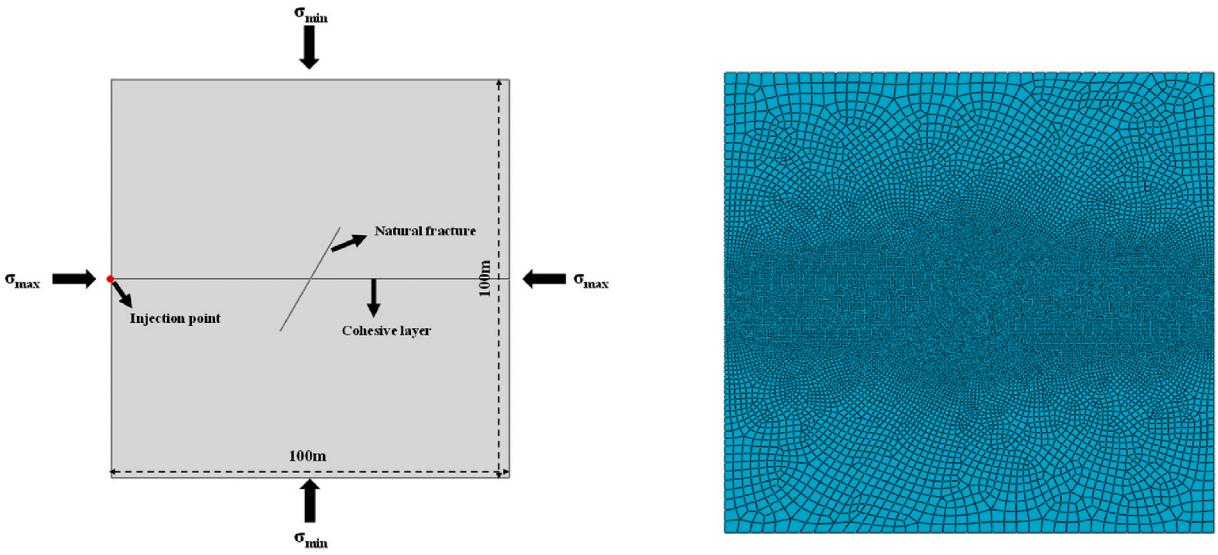


Fig. 1. Two-dimensional model with a single natural fracture.

Table 1
Basic model parameters.

Parameters	Values	Unit
Young's modulus	15/20/30 (rock element) 20 (cohesive element)	GPa
Poisson's ratio	0.25	/
Permeability coefficient	1.00E-07	m/s
Porosity	0.1	/
Leak coefficient	1.00E-14	m/Pa·s
Fluid viscosity	0.001	mP·s
Normal nominal stress	2 (nature fractures) 6 (hydraulic fractures)	MPa
First shear nominal stress	10 (nature fractures) 20 (hydraulic fractures)	MPa
Second shear nominal stress	10 (nature fractures) 20 (hydraulic fractures)	MPa
Maximum principal stress	8/10/13	MPa
Minimum principal stress	8/8/8	MPa
Fluid density	1	g/cm^3
Initial pore pressure	20	MPa
Injection rate	0.001	m^3/s
Injection time	50	s

software ABAQUS, as shown in Fig. 1. The size of the model is 100 × 100 m. The approach angle between natural fracture (NF) and hydraulic fracture (HF) is 60°. The model including 17,153 CPE4P elements and 263 COH2D4P elements to represent rock matrix and fracture respectively. The direction of the maximum principal stress is parallel to the direction of the pre-existing fracture. And the parameters of rock matrix and fractures for this study are shown in Table 1.

2.2. Wavelet analysis model

Wavelet analysis is called "magnifying glass". Comparing with Fourier transform, it has good time-frequency analysis capabilities. Unlike other signal processing methods, the advantage of wavelet analysis is that, it can reveal some hidden details, such as trends, breakdown, and discontinuities in higher-order derivatives (Unal et al., 2019a). It realizes the multi-scale decomposition of the signal through the translate and scale of the wavelet basis. The discrete wavelet is selected to analysis the pressure signal in this study, and the expressions is described by Eq (7) and Eq (8).

$$W(j,k)=\sum_{R} f(n)\frac{1}{\sqrt{a}}\psi_{j,k}(n) \tag{7}$$

$$\psi_{j,k}=\frac{1}{\sqrt{2}}\Psi\left(\frac{t-k2^j}{2^j}\right) \tag{8}$$

Where$f(n)$is the original signal; $\psi_{j,k}$is the wavelet basis function.

The principle of discrete wavelet is to continuously decompose the signal through a series of high-pass and low-pass filters to obtain the high-frequency and low-frequency components of the signal. The high-frequency component (detailed information) of the signal represents discontinuities and singularities, while the low-frequency component represents a rough approximation of the original signal (Awad et al., 2020; Unal et al., 2019a). The detail coefficient and approximate coefficient can be calculated by Eq (9) and Eq (10), and the maximum theoretical decomposition level of the signal can be calculated by Eq (11) (Mallat, 1989).

$$cA_j=\sum_{n=0}^{\infty} f(n)\varphi_{j,k}(n)=\sum_{n=0}^{\infty} f(n)\frac{1}{\sqrt{2^j}}\varphi\left(\frac{n-k2^j}{2^j}\right) \tag{9}$$

$$cD_j=\sum_{n=0}^{\infty} f(n)\psi_{j,k}(n)=\sum_{n=0}^{\infty} f(n)\frac{1}{\sqrt{2^j}}\psi\left(\frac{n-k2^j}{2^j}\right) \tag{10}$$

$$M=\log_2 N \tag{11}$$

Where$f(n)$is the original signal; 2^j、$k2^j$、$j$are the scale parameter, translation parameter and decomposition level, respectively; N represents the signal length.

According to Parseval, the total energy of the signal can be described by Eq (12), and the detailed energy and approximate energy of the signal can be described by Eq (13) and Eq (14) (Burrus et al., 1998; Unal et al., 2019a):

$$E=\sum_{i=0}^{N-1}\left|a_{ij}\right|^2, i=1,2,\ldots,l \tag{12}$$

$$E_i^d=\sum_{j=1}^{M}\left|d_{ij}\right|^2 \tag{13}$$

$$E_l^a=\sum_{j=1}^{M}\left|a_{lj}\right|^2 \tag{14}$$

Where$i$is the wavelet-based decomposition level and$M$is the number of detail-approximation coefficients at each decomposition level.

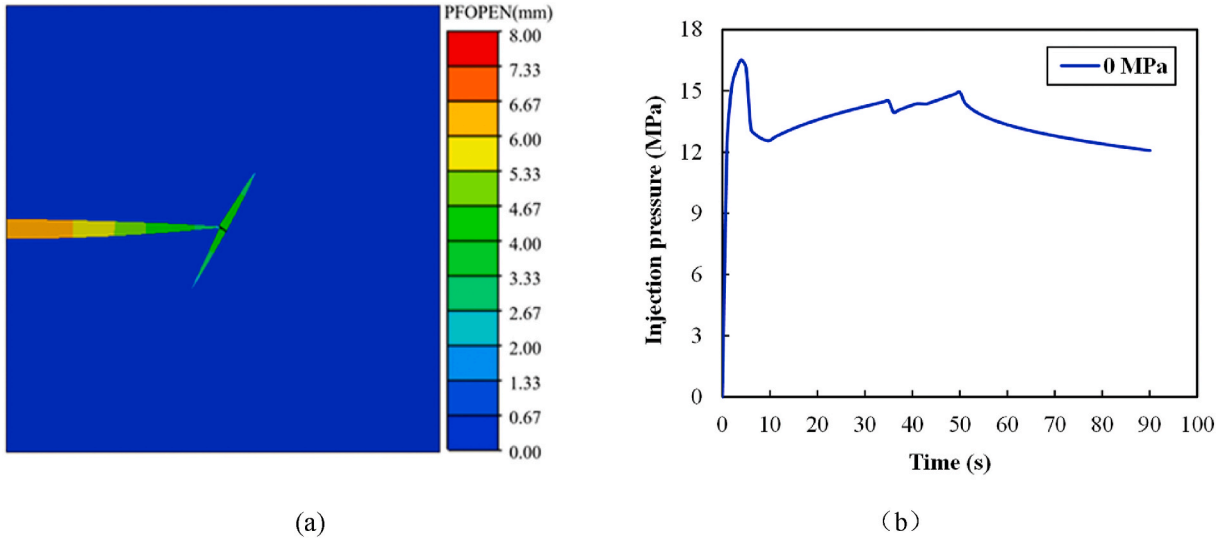


Fig. 2. (a) Hydraulic fracture propagation model under 0 MPa stress difference; (b) injection pressure curve.

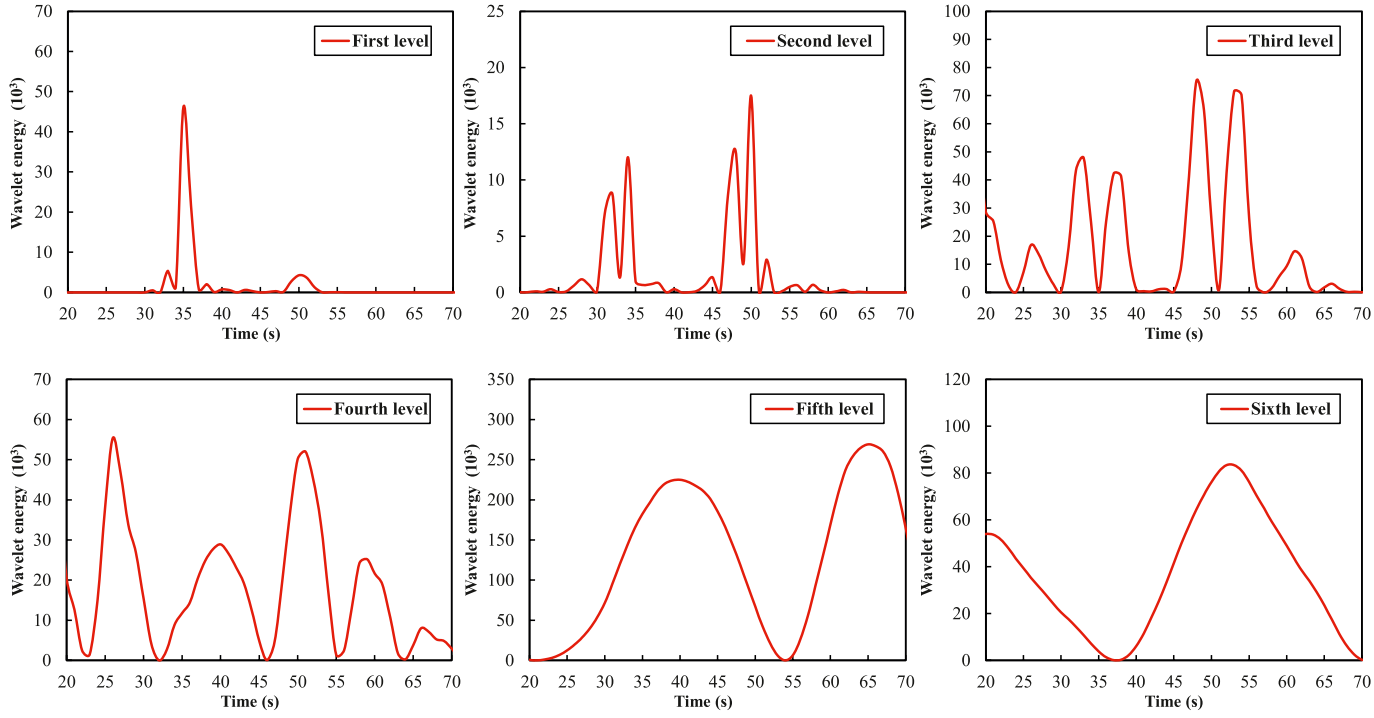


Fig. 3. The wavelet energy at different decomposition levels.

3. Results and discussion

3.1. Effects of wavelet decomposition level

The injection pressure curve at 0 MPa of stress difference is used for a basic case to discuss the recognition result and the optimal recognition level. Fig. 2(a) shows the fracture morphology after hydraulic fracturing under this condition, the deflect of hydraulic fracture is not affected by the stress difference. After encountering natural fracture, hydraulic fracture firstly activates low-angle natural fracture, and then high-angle natural fracture. Fig. 2(b) shows the injection pressure curve at 0 MPa of stress difference. After the initiation stage, there are two obvious pressure drops in the pressure curve, corresponding to the HF and NF encounter event (36s) and the shut-in event (50s). Moreover, the two events are used to wavelet analysis to observe their wavelet response characteristics later. The maximum decomposition level is 6 levels that calculate by Eq (11). Firstly, we use the Db5 wavelet basis to decompose the injection pressure curve in 6 levels, and then the wavelet energy is calculated by Eq (13). Fig. 3 shows the relationship of wavelet energy and time under different decomposition levels. Since we only focus on the response characteristics in the propagation stage of hydraulic fracture, only 20–70s wavelet energy results are shown. During propagation of hydraulic fracture, different decomposition levels represent different wavelet response characteristics.

To quantitatively describe the event recognition result and the optimal recognition level in the wavelet energy response under different decomposition levels, the ratio of wavelet energy variance is defined in the network anomaly detection by Xiao (2006) , is given by Eq (15).

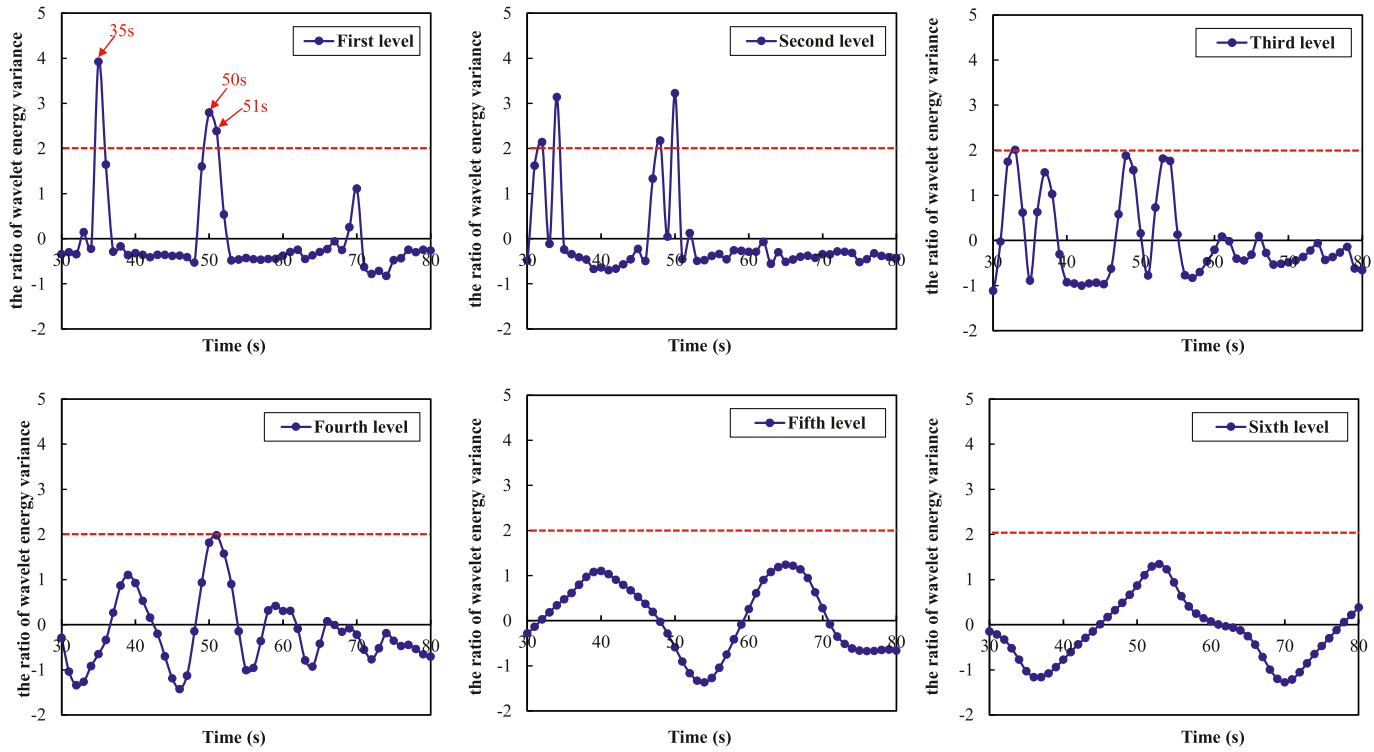


Fig. 4. The ratio of wavelet variance at different decomposition levels.

Generally, we can quantitatively judge whether an abnormality has occurred by comparing the deviation of the wavelet energy from the mean value in a certain time window (the P value is small when the flow is normal, generally less than 1; if the P value exceeds 2, it is considered that an abnormal flow has occurred or ended). During hydraulic fracturing, the normal propagation of hydraulic fracture represents normal flow, and the occurrence of abnormal events represents abnormal flow or end. Fig. 4 shows the ratio of wavelet energy variance between 30s and 80s that calculated with a 20s-time window. Comparing with the injection pressure curve and the numerical simulation results, there are three positions where the ratio of wavelet energy variance is greater than 2 in the first-level wavelet decomposition level, 35s, 50s, and 51s respectively. The value at 35s corresponds to the first wavelet peak energy, it can represent the encounter of HF and NF, while the value of 50s and 51s correspond to the second wavelet peak energy, which can represent the pump shut-in event. The reason for the two identified positions in the pump shut-in event, it may be related to the type of wavelet basis. Therefore, at the first level of decomposition, the peak wavelet energy can be used to reflect abnormal fluctuations on the pressure curve. For other decomposition levels, we have not observed that it is directly related to the abnormal fluctuations on the injection pressure curve.

$$P = \frac{|x_i - \bar{x}|}{S} \tag{16}$$

Which, $x_i$is the sample point, $\bar{x}$ is the mean of the sample value, and$S$ is the variance of the sample value.

3.2. Effects of wavelet basis

The type of wavelet basis is the key to wavelet analysis, is directly related to the signal decomposition result. However, there is still no uniform theoretical standard for the selection of wavelet basis, and the selection is generally based on the purpose of signal processing. Different wavelet basis has different characteristics, the vanishing moment and support length are the key factors that affect the decomposition effect. The vanishing moment of wavelet is mainly used to detect the discontinuous signal of higher order derivative. The larger the vanishing moment, the better the smoothness and the stronger the localization ability in the frequency domain, it can be described by Eq (16). The larger the vanishing moment, the larger the supporting length of the wavelet basis, which increases the amount of calculation. In order to balance the advantages between them, and select the optimal wavelet basis that suitable for the pressure signal in hydraulic fracturing, we compare the decomposition results of three different wavelet basis under different vanishing moments (Db5, Db7, Db9, Db11, Sym 2, Sym4, Sym 6, Sym 8, Coif2, Coif 3, Coif 4, Coif 5)

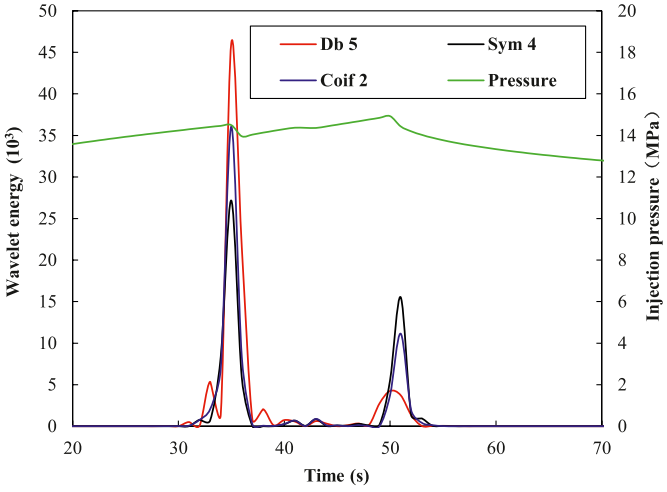


Fig. 5. Wavelet energy response of different wavelet bases under different vanishing moments.

$$\int_{-\infty}^{+\infty} t^l \psi(t) dt = 0 \tag{15}$$

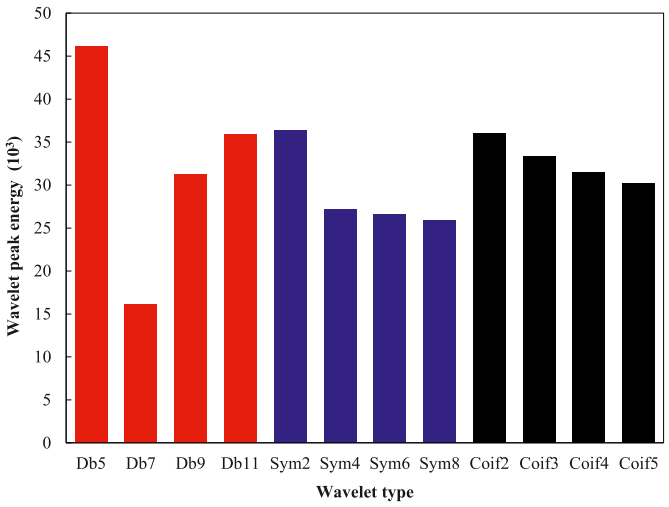


Fig. 6. Wavelet peak energy of different wavelet bases when HF encounter NF.

Where the wavelet functionψsatisfies the above equation for all is said to have M-order vanishing moments and a support length of 2M-1.

Fig. 5 shows the curve of wavelet energy and pressure over time under three different types of wavelet basis (Db5, Sym4, Coif2 wavelet basis). There are two abnormalities in the first-level wavelet energy curve under different wavelet basis, but the wavelet peak energy exists different. As discussed earlier, the two wavelet peak energy correspond to the event of HF and NF encounter and pump shut-in during hydraulic fracture propagation process. Subsequently, we compare the wavelet peak energy of 12 different wavelet basis under the event of HF and NF encounter. The wavelet peak energy of Db5 wavelet basis is the largest, as shown in Fig. 6. Generally, larger peak energy can help us maintain our ability to recognize weak pressure changes, therefore the Db5 wavelet basis is selected as the optimal wavelet basis of injection pressure signal and conducts subsequent analysis.

3.3. *Effects of stress difference*

To investigate the effects of different stress differences on wavelet response, the fracture propagation models under different stress differences (0 MPa, 2 MPa and 5 MPa) are established for wavelet analysis. Fig. 7 shows the fracture propagation morphology under different stress differences. At the stress difference of 0 MPa, the hydraulic fracture first activated low-angle natural fracture and then further activated high-angle natural fracture. At the stress difference of 2 MPa, only low-angle natural fracture was activated. At the stress difference of 5 MPa, the fracture did not activate the natural fracture, but directly pass through the natural fracture.

Fig. 8(a) shows the curve of injection pressure under different stress differences. Before encountering natural fracture, the injection pressure maintains the same variation under different stress differences. Fig. 8(b) shows the curve of wavelet energy under different stress differences. Under different stress differences, two obvious abnormal fluctuations were observed on the wavelet energy curve of first decomposition level. As discussed in Section 3.1, the wavelet peak energy corresponds to the

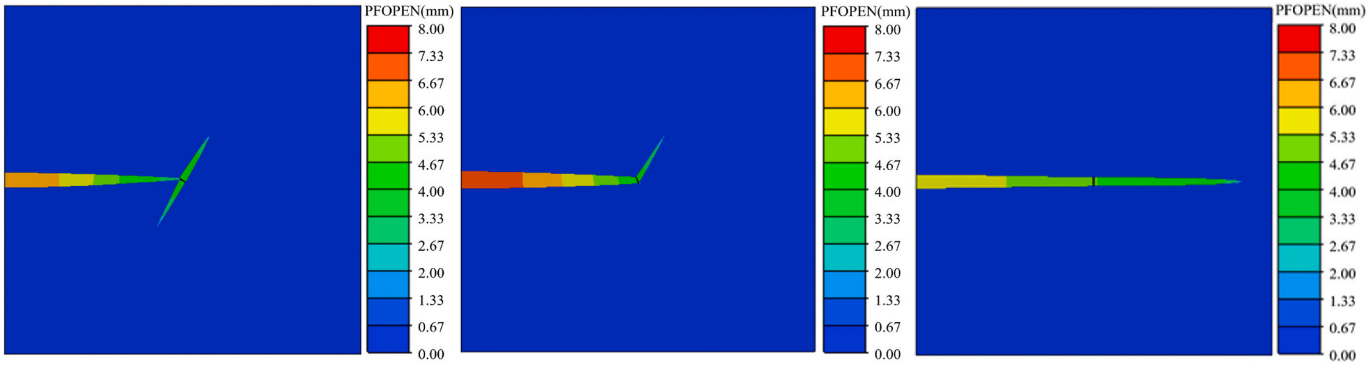


Fig. 7. Propagation model of HF and NF under different stress differences.

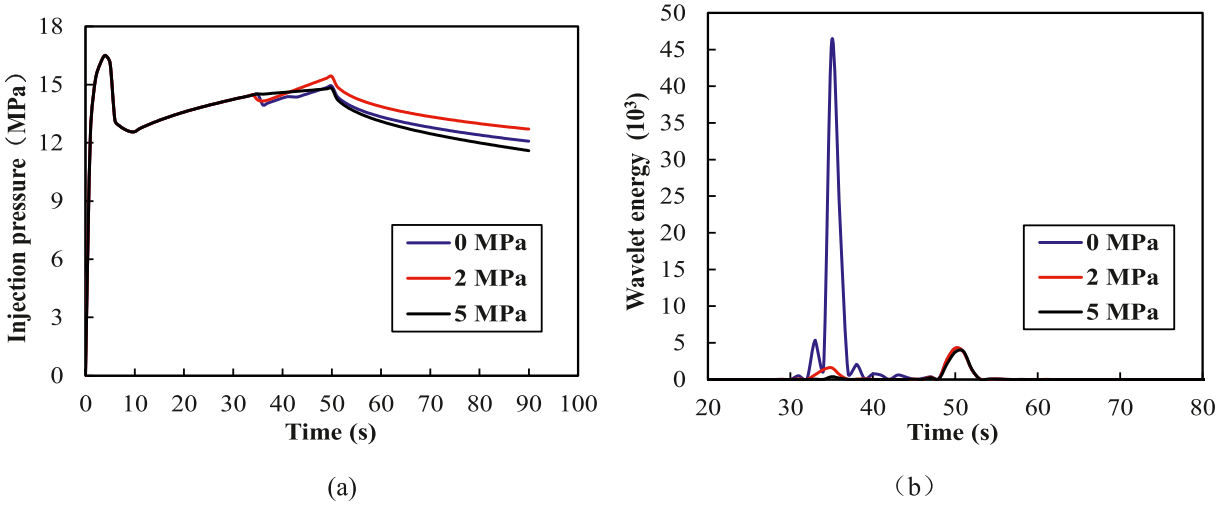


Fig. 8. (a): The injection pressure curve; (b): The wavelet energy curve under different stress difference.

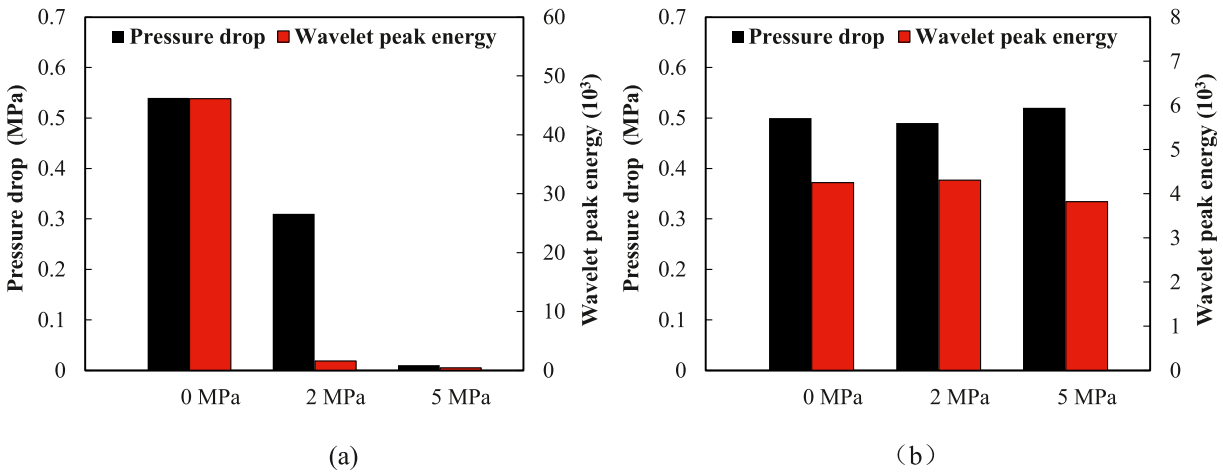


Fig. 9. Pressure drop and wavelet peak energy at different stress differences, (a): When HF encounter NF; (b): When pump shut-in.

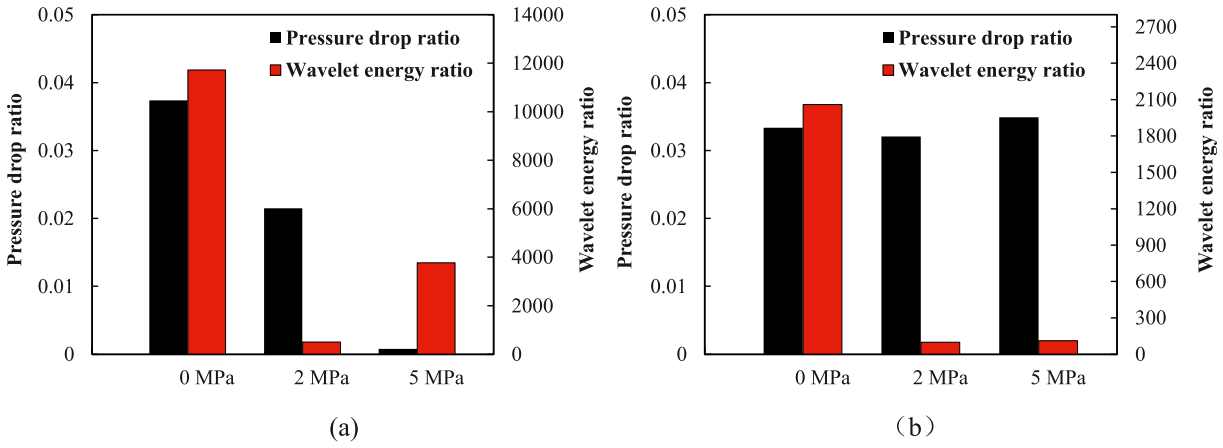


Fig. 10. Pressure drop ratio and wavelet energy ratio at different stress differences, (a): When HF encounter NF; (b): When pump shut-in.

abnormal fluctuation on the pressure curve. Fig. 9 shows the comparison of the pressure drop values and the wavelet peak energy of the two abnormal events under different stress differences.

At the moment that encounter of HF and NF, when the stress difference is 0 MPa, the pressure drop is 0.54 MPa; when the stress difference is 2 MPa, the pressure drop is 0.31 MPa; when the stress difference is 5 MPa, the pressure drop is 0.01 MPa. Hence, with the stress difference increases, the pressure drop gradually decreases. Under different stress differences, the wavelet peak energy is 46.12, 1.59, and 0.39 ($\times 10^3$) respectively. With the stress difference increases, the wavelet peak energy decreases. For the shut-in event, the pressure drop values under different pressure difference are 0.50 MPa, 0.49 MPa, and 0.52 MPa, and the wavelet peak energy is 4.25, 4.31, and 3.82 ($\times 10^3$), respectively. The difference between the pressure drop value and the wavelet peak energy is small under different stress difference. Therefore, the difference in the pressure drop value is mainly caused by the stress difference resulting in the appearance of multiple interaction modes between hydraulic and natural fractures. Meanwhile, there is a certain positive correlation between this pressure drop value and the wavelet peak energy. Simultaneously, a positive relationship was found between the magnitude of wavelet peak energy and the value of pressure drop.

The pressure drop ratio and the wavelet energy ratio is defined to

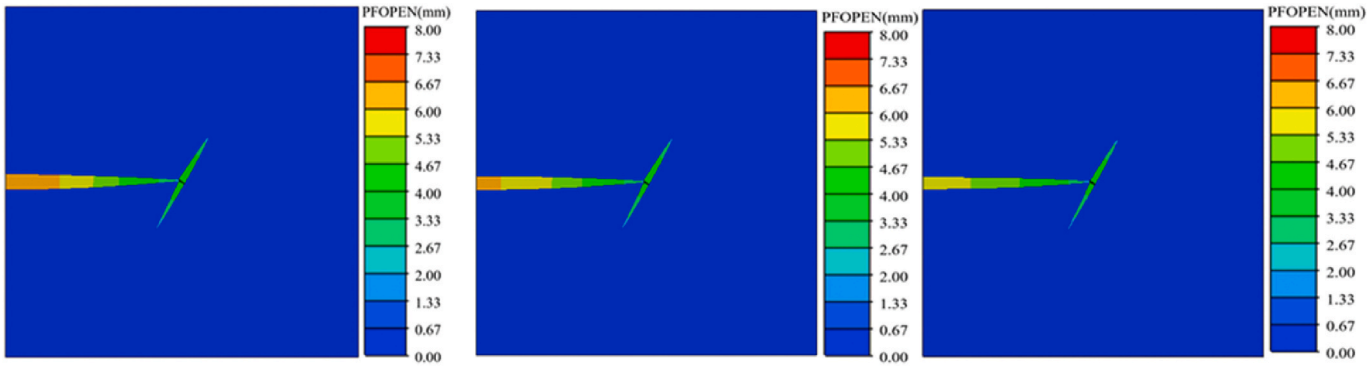


Fig. 11. Propagation model of HF and NF under different elasticity modulus.

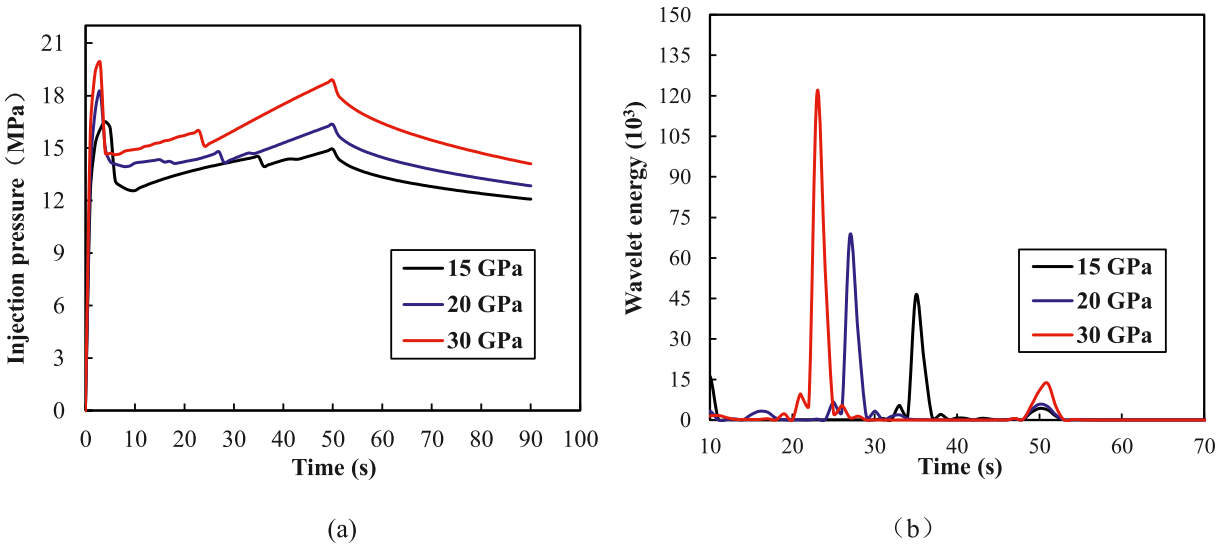


Fig. 12. (a): The injection pressure curve; (b): The wavelet energy curve under different elasticity modulus.

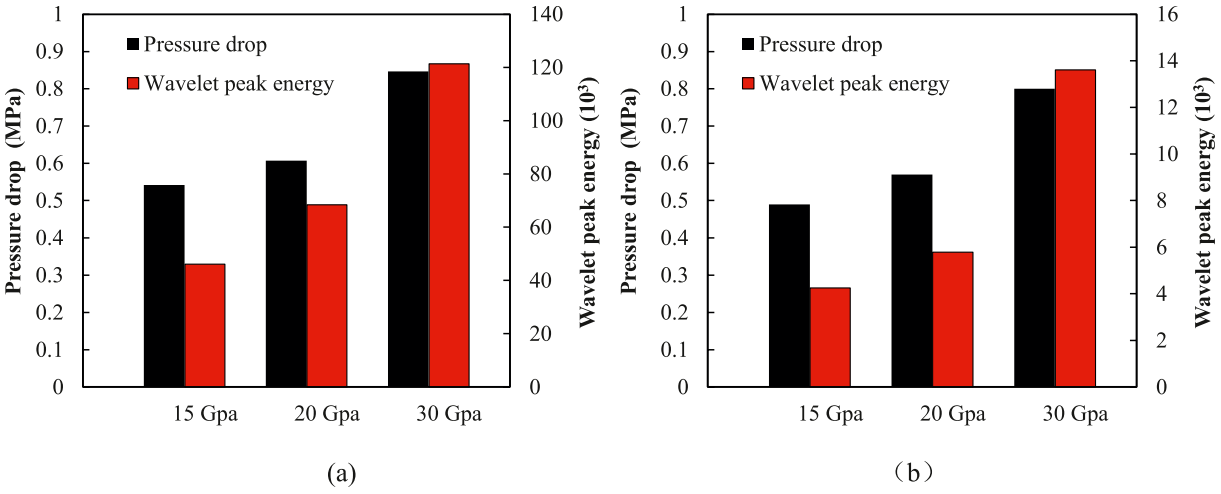


Fig. 13. Pressure drop and wavelet peak energy at different elasticity modulus, (a): When HF encounter NF; (b): When pump shut-in.

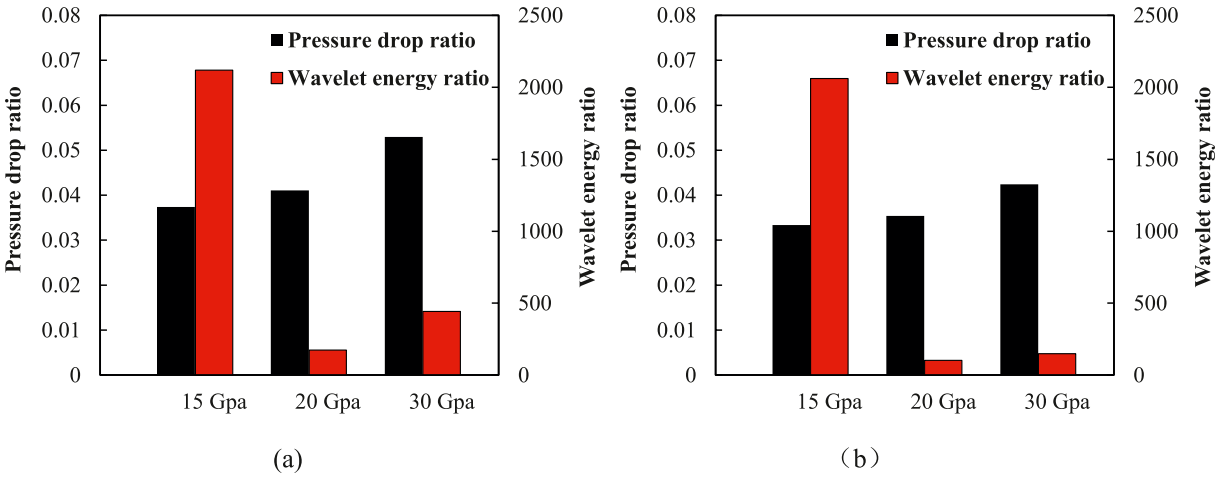


Fig. 14. Pressure drop ratio and wavelet energy ratio at different stress differences, (a): When HF encounter NF; (b): When pump shut-in.

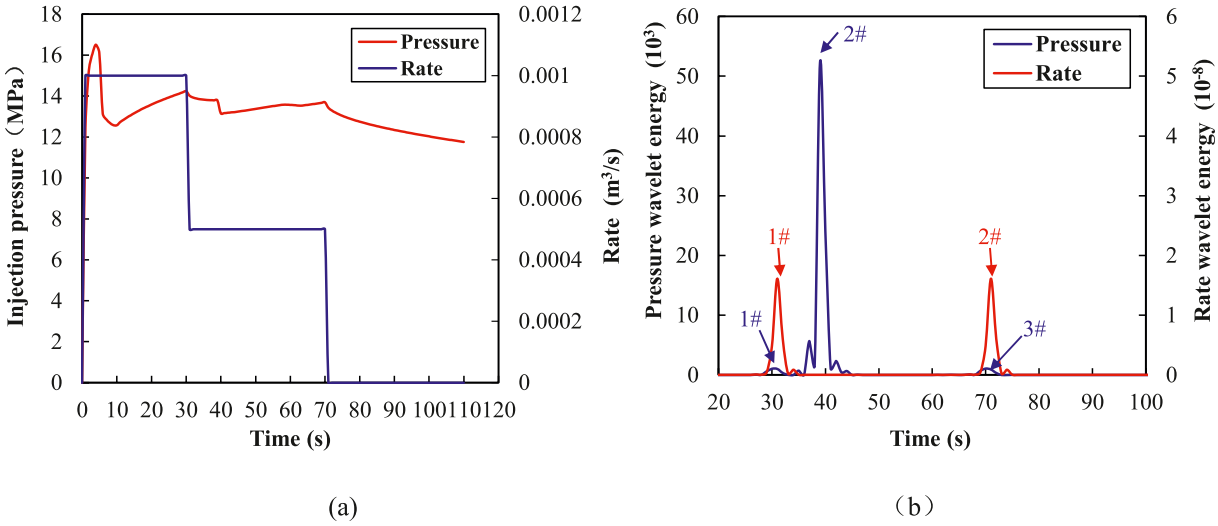


Fig. 15. (a): The injection pressure and rate curve; (b): The wavelet energy curve of pressure and rate.

compare the difference of the pressure fluctuation response and the wavelet energy response. The pressure drop ratio is defined as the ratio of the pressure drop of the abnormal event to the injection pressure, and the wavelet energy ratio is defined as the ratio of the peak energy of the abnormal event to the wavelet energy of the normal fracture propagation stage. Fig. 10 shows the comparison of the pressure drop ratio and the wavelet energy ratio of two abnormal events under different stress difference conditions. The pressure drop ratios are 0.037、0.022 and 0.001 when HF encounter NF, the energy ratios are 11723.131, 503.095, and 3769.813 respectively. The difference between them is 4–6 orders of magnitude. Therefore, the wavelet response of pressure can greatly amplify the abnormal fluctuations on the pressure curve. Thus provide a basis for further analysis of the event.

3.4. *Effects of elastic modulus*

To investigate the effects of different elastic modulus on pressure fluctuation and wavelet response, the fracture propagation model is established under different elastic modulus (15, 20, 30 GPa). Fig. 11 shows the different fracture morphology at different elastic modulus models. The elastic modulus has little effect on the propagation of hydraulic fractures. During the hydraulic fracture propagation process, the low-angle natural fracture first is activated after HF encounter NF, and then along the high-angle natural fracture to propagation.

Fig. 12(a) shows the curve of injection pressure under different elastic modulus (15, 20, 30 GPa). With the elastic modulus increasing, the breaking pressure gradually increases. During the hydraulic fracture propagation stage, the injection pressure curve has two pressure drops correspond to the event of HF encounter NF and the event of pump shut-in. Fig. 12(b) shows the curve of wavelet energy under different elastic modulus. The curve demonstrates that the wavelet energy under different elastic modulus has two obvious abnormal peaks, but the wavelet peak energy exist difference. Fig. 13 shows the comparison of the pressure drop value and wavelet peak energy under different elastic modulus. The pressure drop value is 0.54 MPa and 0.61 MPa、0.85 MPa respectively when HF encounter NF, the wavelet peak energy is 46.12, 68.45, 121.36 respectively, with the elastic modulus increasing, the pressure drop value and the wavelet peak energy both increase. The variation patterns of the pressure drop values and wavelet peak energy for the pump shut-in event remain the same as the HF and NF encounter event. Therefore, there is a positive correlation between the pressure drop value and the wavelet peak energy.

Fig. 14 shows the pressure drop ratio versus wavelet energy ratio for

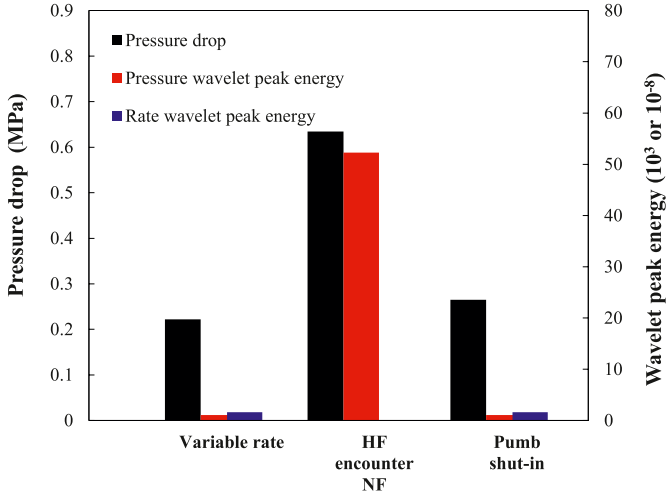


Fig. 16. Pressure drop and wavelet peak energy under variable rate conditions.

the two anomalous events under different elastic modulus. When HF encounter NF, the pressure drop ratio lies between 0.04 and 0.05, which is difficult for the naked eye to observe the abnormal pressure fluctuation. However, the wavelet energy ratio lies between 174.78 and 2119.14 on the curve of wavelet energy, which is 4–5 orders of magnitude larger than the pressure drop ratio. Therefore, it is convenience to use wavelet analysis to capture abnormal fluctuations on the injection pressure curve.

3.5. *Effects of variable rate*

Pressure fluctuations in the hydraulic fracturing process are affected by the combination of factors (geological factors, treatment factors, etc.). Previously, we discussed the pressure response and wavelet energy response due to geological factors. Next, the sudden change in rate is considered during hydraulic fracturing. And the fracture propagation model with the sudden in rate (reduced by 50% at 30s) is built, the curve of injection pressure and rate is shown in Fig. 15(a). There are three obvious pressure drops in the pressure curve, corresponding to the three events in the hydraulic fracturing process set in this paper, variable rate event, HF encounter NF event, and pump shut-in event, respectively. Wavelet analysis is performed for injection pressure and rate, Fig. 15(b)

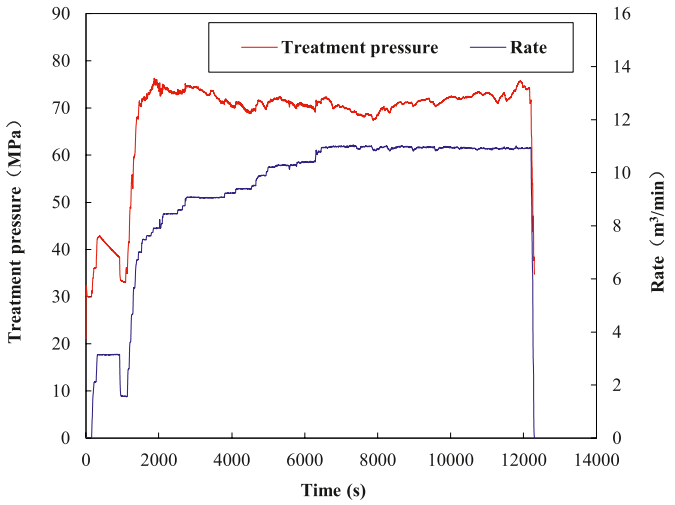


Fig. 17. Field treatment curve of a well.

shows the wavelet energy response law for both. There are three obvious abnormal peaks in the pressure wavelet energy, which correspond to the variable rate event, HF encounter NF event, and the pump shut-in event, respectively. And there are two obvious abnormal peaks on the wavelet energy curve, which correspond to the variable rate event and the pump shut-in event. Simultaneous, it can be found that the 1# peak and 3# peak on the injection pressure curve correspond with the two peaks (1# and 2#, red symbols) on the rate curve.

Fig. 16 shows the drop values and wavelet peak energy of pressure and rate under different abnormal events. The pressure drop values are 0.23, 0.63, 0.26 MPa, respectively. The pressure drop caused by the encounter of HF and NF is relatively large, while the pressure drop value is small that caused by the event of variable rate and pump shut-in. The wavelet peak energy is 1.03, 52.30, 1.06 ($\times 10^3$) respectively, which is directly related to the value of pressure drop. The sudden change of the rate and the pump shut-in event both can be regarded as a 50% reduction in the rate. Therefore, the wavelet peak energy is equal to 1.61, demonstrated that there is a direct correlation between the drop value and the wavelet peak energy. Furthermore, rate sudden changes not only can cause fluctuations on the pressure wavelet energy curve, but also cause some response on the rate wavelet energy curve. Comparing the differences between them can provide more intuitive guidance for analyzing rock events and fluid events in fracturing process.

4. Field case

The field hydraulic fracturing treatment curve is used to wavelet analysis to discuss the response law of treatment pressure and rate. As shown in Fig. 17, the hydraulic fracturing lasted 12,300 s in total and the sampling frequency was 1 Hz. The data from 2000s to 12000s is selected for wavelet analysis by Db7 wavelet basis to sure the stability of analysis. And the maximum decomposition level is 13 levels by Eq (11). The specific operation detail is: firstly, the treatment pressure curve and rate curve are decomposed by Db7 wavelet base. Then, the wavelet energy response of construction pressure and displacement respectively by Eq (13). Finally, the hydraulic fracturing events are divided into rate events and non-rate events by comparing the differences between them.

Fig. 18 shows the wavelet energy of the treatment pressure and rate at the 7th level of decomposition. Since the data boundary will have a great influence on the wavelet decomposition, there only shows the wavelet decomposition effect of 3500–11000s. It can be seen from the figure that the wavelet energy around 3867, 4639, 5020, 7575, 8987, 9621, 10,272, 10658s shows that the rate wavelet response is not obvious, but the pressure response is very severe. These events are

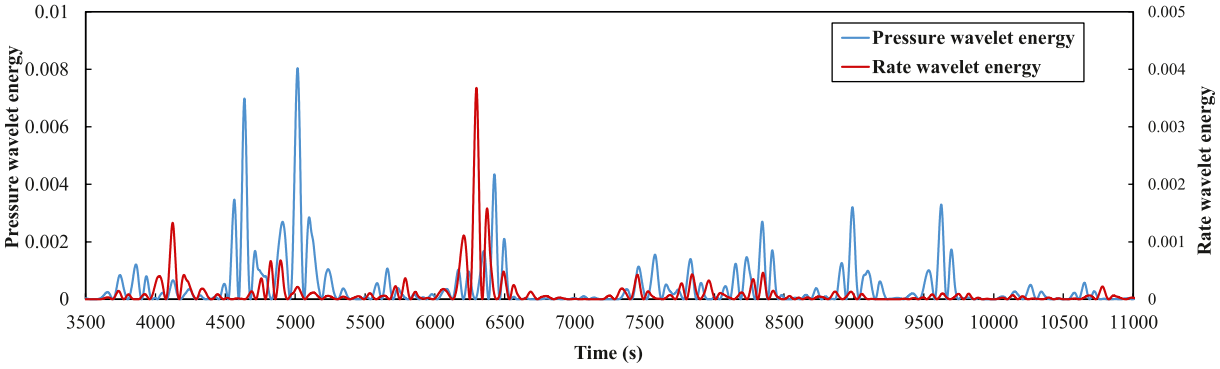


Fig. 18. Pressure and rate wavelet energy versus time curve at the 7th level of decomposition.

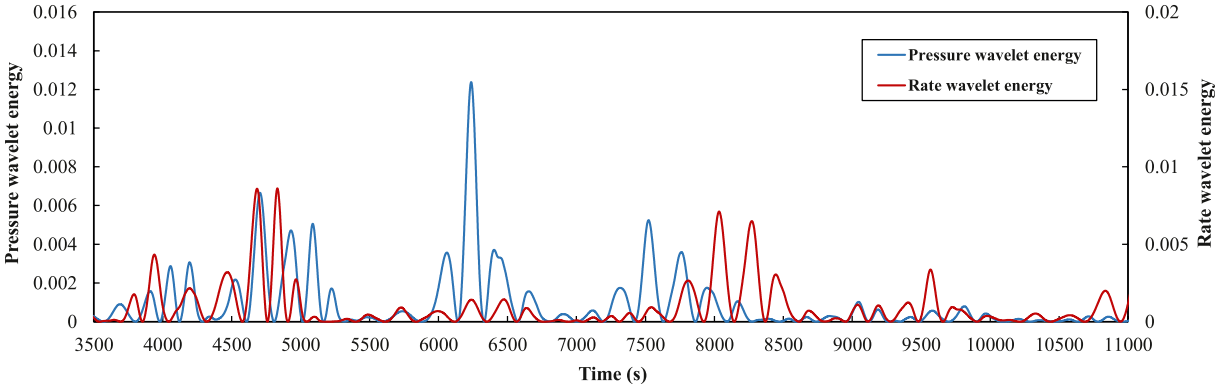


Fig. 19. Pressure and rate wavelet energy versus time curve at the 8th level of decomposition.

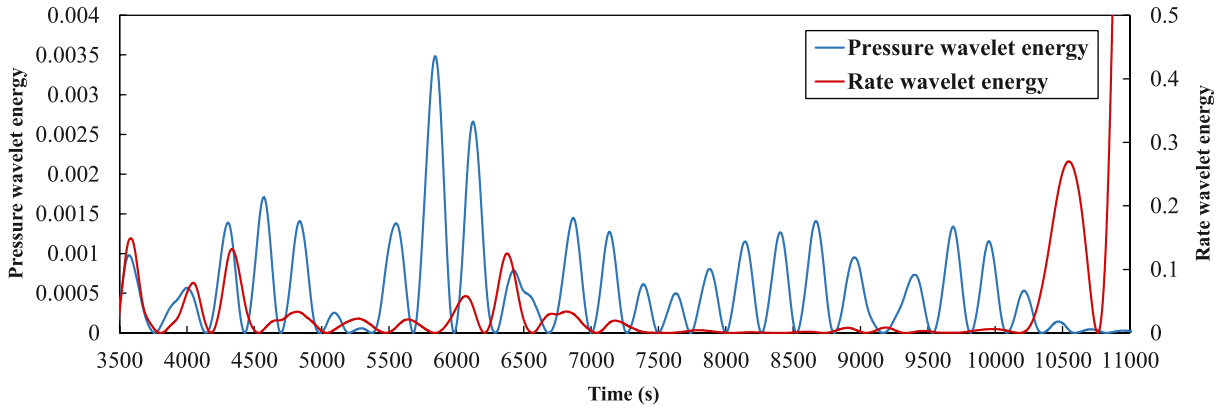


Fig. 20. Pressure and rate wavelet energy versus time curve at the 9th level of decomposition.

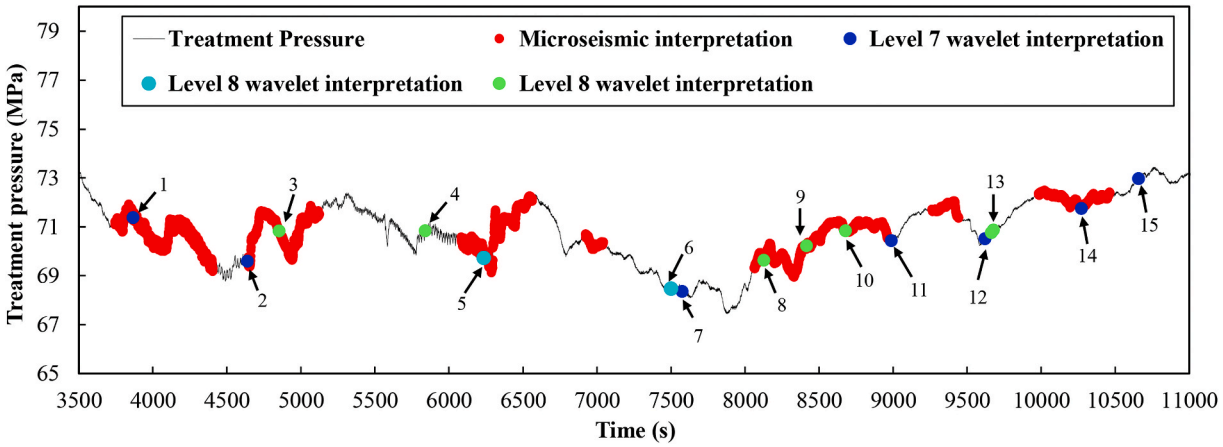


Fig. 21. Comparison of microseismic interpretation results and wavelet analysis results.

considered as NON-RATE EVENT, represented many rock events such as the fracture propagation, the interaction of natural fractures, and so on. On the contrary, the wavelet energy response of pressure and rate has good consistency, which shows that the pressure response at these times is likely to be caused by the fluctuation of rate, so it is considered as RATE-EVENT. Similarly, at the 8th level of decomposition, the rate response at moments around 6236 and 7500 s is not obvious, but the pressure response is very serve, that is, the NON-RATE EVENT (Fig. 19). At the 9th level of decomposition, at 4852, 5839, 8415, 8679, 9677, 9664s, there are NON-RATE EVENT events (Fig. 20). In order to verify the rationality of wavelet analysis, the results of wavelet analysis are compared with micro-seismic interpretation, as shown in Fig. 21. The wavelet interpretation results obtained a total of 15 event points. The red dots represent the microseismic interpretation results, and the blue dots represent the 7 h-level wavelet energy interpretation results (1, 2, 7, 11, 14, 15), the sky blue represents the 8th level wavelet energy interpretation results (5, 7), and the green dots represent the 9th level wavelet energy interpretation results (3, 4, 8, 9, 10, 13). Except for event points 4, 6, 7, 12, 13, and 15, all other event points are in the microseismic interpretation results.

5. Conclusion

(1) Different wavelet decomposition levels can reflect the performance of the signal in different frequency bands. For the hydraulic fracture propagation model in this study, the maximum decomposition level of the pressure signal is 6 levels, and there are two anomalies peak in the first decomposition level during hydraulic fracture propagation stage, better correspond to the hydraulic fracture and natural fracture encounter event and the pumping shut-in event. Comparing the decomposition effects of three different wavelet bases under different vanishing moments, it is found that Db5 is the optimal wavelet base for the injection pressure signal.

(2) There is a positive correlation between the pressure drop value and the wavelet peak energy. With the increase of stress difference, the pressure drop and wavelet energy both decreases (HF encounter NF). With the elasticity modulus increases, the pressure drop and wavelet energy both increases. Comparing the wavelet energy ratio and pressure drop ratio can reveals that the wavelet analysis method can greatly amplify the response to pressure fluctuations caused by abnormal events.

(3) Comparing the wavelet energy of pressure and rate can provide a certain reference for the event analysis of hydraulic fracturing. In order to prove the feasibility of the method, the results of microseismic interpretation were used for comparison.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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